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Course Code	Lab 4_SBT_DF203
Lab Title	SMTP Email Traffic Forensics
Recommended Environment	Windows 10/11 host, Python ,CMD and Wireshark
Submission Format	One PDF report plus the created .bat files and-output evidence files
Course Code	SBT_DF203
Instrutor name	Aminu Idris AMCPN
Date	14 September, 2026

Step 1

smtp.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.10.1.4	10.10.1.1	DNS	76	Standard query 0x7956 A mail.patriots.in
2	0.034025	10.10.1.1	10.10.1.4	DNS	142	Standard query response 0x7956 A mail.patriots.in CNAME patriots.in A 74.53.140.153 NS ns2.p
3	0.036986	10.10.1.4	74.53.140.153	TCP	62	1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM
4	0.383936	74.53.140.153	10.10.1.4	TCP	62	25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM
5	0.383968	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.727683	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 We do n
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800 PIPELINING AUT
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 VXN1cm5hbWU6
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycGFydGFiQHBhdHJpb3RzImlu
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 UGFzc3dvcmQ6
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamFiQDEyWm==

< Frame 1: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
> Ethernet II, Src: CradlePoint_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:0c:29:81:60:00)
> Internet Protocol Version 4, Src: 10.10.1.4, Dst: 10.10.1.1
> User Datagram Protocol, Src Port: 56166, Dst Port: 53
> Domain Name System (query)

0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00 --3...<...E-
0010 00 3e 25 0a 00 00 00 11 ff 8c 0a 0a 01 04 0a 0a -->%...<...J5
0020 01 01 db 66 00 35 00 2a 8e 8d 79 56 01 00 00 01 --f.5.*...b P-
0030 00 00 00 00 00 04 6d 61 69 6c 08 70 61 74 72 --h...AU TH LOGIN
0040 69 6f 74 73 02 69 6e 00 00 01 00 01 ..

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AUTH/LOGIN

smtp.pcap

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No.	Time	Source	Destination	Protocol	Length	Info
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800 PIPELINING AUT
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 VXN1cm5hbWU6
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycGFydGFiQHBhdHJpb3RzImlu
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 UGFzc3dvcmQ6
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamFiQDEyWm==
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted
20	2.828143	10.10.1.4	74.53.140.153	SMTP	60	C: DATA

< Frame 10: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
> Ethernet II, Src: CradlePoint_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:0c:29:81:60:00)
> Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153
> Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 10, Ack: 319, Len: 1
> Simple Mail Transfer Protocol

0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00 --3...<...E-
0010 00 34 25 24 40 00 00 06 f3 c3 0a 0a 01 04 4a 35 --4%\$@...<...J5
0020 8c 99 05 be 00 19 7e c4 53 ba ae ec 62 ee 50 18 --h...AU TH LOGIN
0030 fe c1 89 68 00 00 41 55 54 48 20 4c 4f 47 49 4e --
0040 0d 0a ..

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USERNAME

smtp.pcap

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No.	Time	Source	Destination	Protocol	Length	Info
60	9.198384	10.10.1.20	10.10.1.255	BROWSER	243	Host Announcement MIT-3D655AA090D, Workstation, Server, NT Workstation, Potential Browser, E
1	0.000000	10.10.1.4	10.10.1.1	DNS	76	Standard query 0x7956 A mail.patriots.in
3	0.836986	10.10.1.4	74.53.140.153	TCP	62	1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM
5	0.383968	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycGFydGFWQHBhdHJpb3RzLm1u
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamF1QDEyYm==
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
20	2.828143	10.10.1.4	74.53.140.153	SMTP	60	C: DATA
22	3.200683	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
23	3.200726	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
24	3.200744	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes

> Frame 12: Packet, 84 bytes on wire (672 bits), 84 bytes captured (672 bits)

> Ethernet II, Src: CradlePoint_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:0c:29:81:60:00)

> Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153

> Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 22, Ack: 337, Len: 84

> Simple Mail Transfer Protocol

0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00 ...3...<...E-
0010 00 46 25 29 40 00 00 06 f3 ac 0a 0a 01 04 4a 35 ...F%...)@...J5
0020 8c 99 05 be 00 19 7e c4 53 c6 ae ec 63 00 50 18S...c P-
0030 fe af 22 a4 00 00 5a 33 56 79 63 47 46 79 64 47Z3 VycGFydG
0040 46 77 51 48 42 68 64 48 4a 70 62 33 52 7a 4c 6d FwQHBhdH Jpb3RzLm
0050 6c 75 0d 0a 1u

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PASSWORD

smtp.pcap

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No.	Time	Source	Destination	Protocol	Length	Info
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 We do n
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800 PIPELINING AUT
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	C: 334 VXN1cm5hbWU6
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycGFydGFWQHBhdHJpb3RzLm1u
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	C: 334 UGFzc3ducmlu
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamF1QDEyYm==
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	C: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted

> Frame 14: Packet, 72 bytes on wire (576 bits), 72 bytes captured (576 bits)

> Ethernet II, Src: CradlePoint_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:0c:29:81:60:00)

> Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153

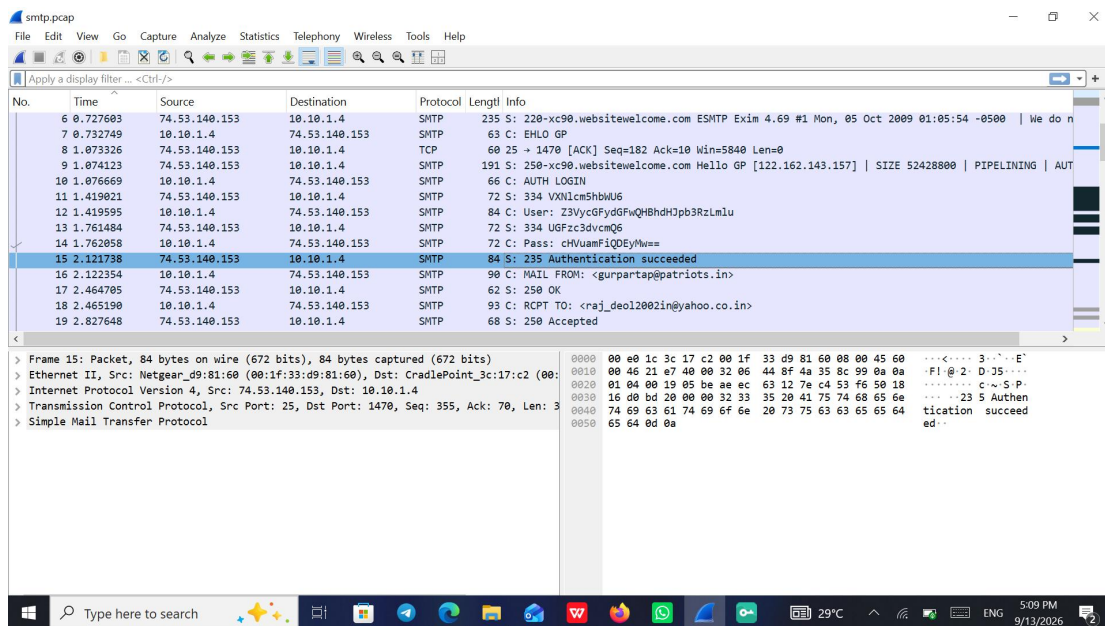
> Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 52, Ack: 355, Len: 1

> Simple Mail Transfer Protocol

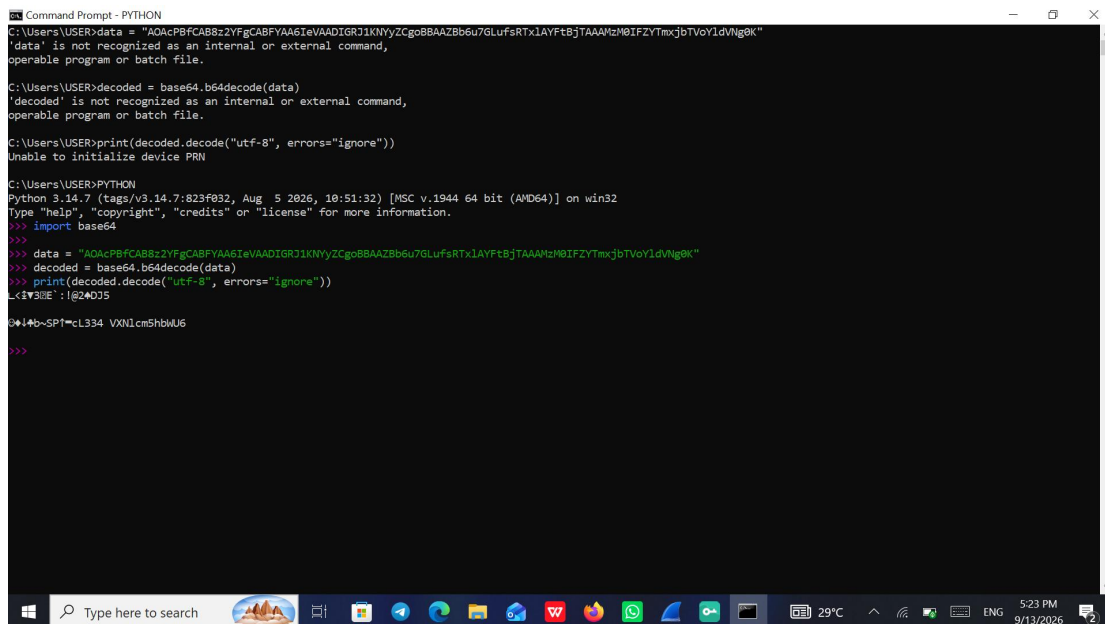
0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00 ...3...<...E-
0010 00 3a 25 2e 40 00 00 06 f3 b3 0a 0a 01 04 4a 35 ...%.@...J5
0020 8c 99 05 be 00 19 7e c4 53 e4 ae ec 63 12 50 18S...c P-
0030 fe 9d 54 01 00 00 63 48 56 75 61 6d 46 69 51 44 ...T...cH VuamF1QD
0040 45 79 4d 77 3d 3d 0d 0a EyWm==...

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235 AUTHENTICATION



DECODE



USER DECODE

```
Command Prompt - PYTHON
C:\Users\USER>data = "AQAcPBFCAB8z2YFgCABFYAA6IeVAADIGR3IKNyZCgoBBAZBb6u7GLufSRTxJAYFtBjTAAAHzM0IFZYTmxjbtVoY1dVNg0K"
'data' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\USER>decoded = base64.b64decode(data)
'decoded' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\USER>print(decoded.decode("utf-8", errors="ignore"))
Unable to initialize device PRN

C:\Users\USER>PYTHON
Python 3.14.7 (tags/v3.14.7:823f032, Aug 5 2026, 10:51:32) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import base64
>>>
>>> data = "AQAcPBFCAB8z2YFgCABFYAA6IeVAADIGR3IKNyZCgoBBAZBb6u7GLufSRTxJAYFtBjTAAAHzM0IFZYTmxjbtVoY1dVNg0K"
>>> decoded = base64.b64decode(data)
>>> print(decoded.decode("utf-8", errors="ignore"))
<Iv3DE'!@2ADJ5

334 VXNlcm5hbmU6

>>> AB8z2YFgAQAcPBFCABFAABG3S1AAIAG86wKCGEESjWmQm+AB1+xFPGruxjAFAY/qBipAAWjMwEwHnRlkR8Z3UUhCaGRISnBm1J6TG1sdQ0K
Traceback (most recent call last):
  File "<python-input-5>", line 1, in <module>
    AB8z2YFgAQAcPBFCABFAABG3S1AAIAG86wKCGEESjWmQm+AB1+xFPGruxjAFAY/qBipAAWjMwEwHnRlkR8Z3UUhCaGRISnBm1J6TG1sdQ0K
NameError: name 'AB8z2YFgAQAcPBFCABFAABG3S1AAIAG86wKCGEESjWmQm' is not defined
>>>
```

```
Wireshark - Follow TCP Stream (tcp.stream eq 0) - smtp.pcap

334 VXNlcm5hbmU6
Z3VycG9ydG9wQHBhdHJp3BzRzLnlu

334 UGFzc3dvcmQ6
cHVuamFiQ0EyWm==

235 Authentication succeeded

MAIL FROM: <gurpartap@patriots.in>

250 OK

RCPT TO: <raj_deol2002in@yahoo.co.in>

250 Accepted

DATA

354 Enter message, ending with "." on a line by itself

From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca45815ef9e57f05cedb07d05@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
boundary="-----NextPart_000_0004_01CA4580_095693F0"
X-Mailer: Microsoft Office Outlook 12.0
Thread-Index: AcpFgen9BVjJZEDeR1Kh8i+hUyVo0A==
Content-Language: en-us

Packet 22: 19 client pkts, 10 server pkts, 18 turns. Click to select.
Entire conversation (15 kB) Show as ASCII No delta times Stream 0
Find: MICROSOFT OFFICE OUTLOOK Find Next
Filter Out This Stream Print Save as... Back Close Help
```

Evidence 9: Email Headers (TCP Stream Output) This show the email metadata captured from the SMTP traffic.

This section of the TCP Stream output reveals the email metadata, also known as headers. These headers provide

forensic evidence about the message, including the sender, recipient, subject, date, unique message identifier, MIME information, and client software used. Capturing these details is essential in digital forensics because they establish the authenticity of the communication, the timeline of transmission, and the software environment in which the email was composed.

These are highlighted points

From →	"Gurpartap Singh" <gurpartap@patriots.in>
To →	<raj_deol2002in@yahoo.co.in>
Subject →	SMTP
Date →	Mon, 5 Oct 2009 11:36:07 +0530
Message-ID →	<000301ca4581\$9e57f0\$cedb070d\$@in>
MIME-Version →	1.0
Content-Type →	multipart/mixed; boundary=...
X-Mailer →	Microsoft Office Outlook 12.0

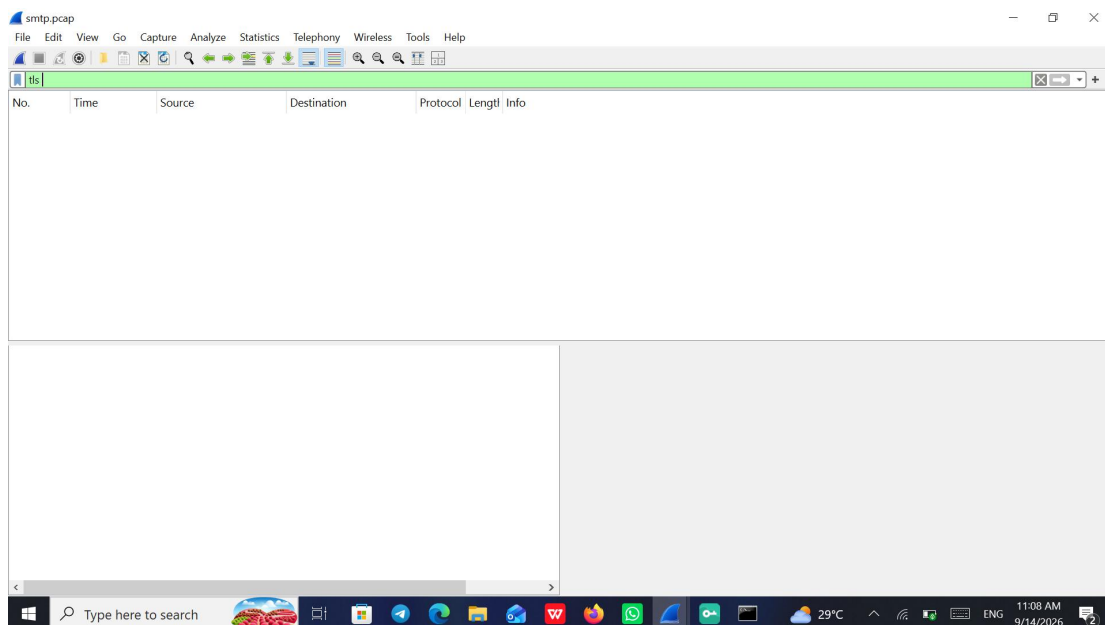
Report Evidence 1: SHA-256 Hash of smtp.pcap.

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>certutil -hashfile C:\Users\USER\Downloads\smtp.pcap SHA256
SHA256 hash of C:\Users\USER\Downloads\smtp.pcap:
17ad238db1b6fd5dd18eb311892df1cf6eb162054bdfb47697b89bef5a86a47ab
CertUtil: -hashfile command completed successfully.

C:\Users\USER>
```

Report evidence 10: STARTTLS/TLS (Not Present).



Details Report

The Wireshark capture shows that no STARTTLS/TLS negotiation occurred during the SMTP session. This indicates the email was transmitted in plaintext, meaning both authentication and message content were visible in the capture. The absence of encryption is

an important forensic observation because it demonstrates how sensitive information can be intercepted when STARTTLS/TLS is not enforced..